Gabriela Mistral:
Wikipedia Entity and Link Analysis

Project Report

Author: Danish Touseeq

A Python NLP pipeline for named entity recognition, outgoing Wikipedia link collection, and candidate person-attribute extraction.

Abstract

This project explores a lightweight natural language processing (NLP) pipeline for analyzing the English Wikipedia article about Gabriela Mistral. It retrieves article text and links, applies spaCy named entity recognition (NER), identifies candidate person names, and exports structured CSV files for inspection and further graph analysis.

1. Objectives

The objectives are to retrieve a Wikipedia biography programmatically, extract named entities, collect outgoing links, represent mentions and links as an edge list, and collect approximate attributes from candidate linked pages.

2. Tools

Python is used for the pipeline and data handling. Wikipedia-API retrieves article text and links. spaCy with the en_core_web_sm model performs NER. pandas exports structured CSV files, and pytest supports tests of utility functions.

3. Methodology

The script retrieves the selected article and normalizes whitespace before processing the text with spaCy. Unique entity spans are saved with their labels. Person-labelled spans are exported as text mentions. Outgoing links are recorded separately; a simple name-pattern heuristic identifies candidate person pages. A bounded number of candidate pages are then retrieved, and approximate birth/death fields are extracted using regular expressions.

4. Output files

The pipeline writes six CSV files: article attributes, named entities, person mentions, outgoing links, person-link edges, and candidate linked-page attributes. Keeping these outputs separate supports manual validation and downstream graph analysis.

5. Results

No live extraction results are claimed in this report because the project was packaged without running against the live Wikipedia API. Counts and CSV contents depend on the article revision and API response at execution time. After running the pipeline, report the actual counts of entities, mentions, links, and successfully retrieved candidate pages.

6. Limitations

spaCy NER identifies likely entity spans but does not resolve identity or prove a relationship. A two-or-more-token name heuristic is not a reliable person classifier. Regex extraction can fail when article wording differs from expected patterns or when a sentence mentions multiple locations. The extracted attributes require validation before use as research evidence.

7. Ethics, provenance, and reproducibility

Wikipedia content changes over time. For reproducibility, record the run date, article URL, and ideally the revision identifier. Respect Wikimedia API guidance and use an informative user-agent with a valid contact method for large-scale access. Review licensing and attribution requirements before redistributing article text or substantial adaptations.

8. Conclusion

The project provides a reproducible starting point for extracting entities and link-based candidate relationships from a Wikipedia biography. The resulting CSV files can support manual validation and further graph analysis, provided heuristic outputs are treated as candidates rather than verified facts.

Running the project

Install the dependencies in requirements.txt, download the spaCy model with `python -m spacy download en_core_web_sm`, and run `python -m src.analyze_wikipedia`. The README contains complete setup instructions.